

1. IEEE RTS-79 System

The IEEE RTS-79 test system consists of 24 buses, 38 transmission lines, and 32 generating units. To represent a power system with renewable energy integration, all 12 MW and 20 MW conventional generators in the original system are replaced by wind farms with the same installed capacities.

For operational reliability evaluation, generator outage states are enumerated up to the fourth order, while transmission-line outage states and combined generation–transmission outage states are enumerated up to the third order, resulting in a complete contingency set.

The day-ahead forecasts of load demand and renewable generation are shown in Fig. 1. The maximum renewable generation output is approximately 40% of the total load demand during the study period. The parameters of conventional generating units in the modified IEEE RTS-79 system are listed in Table 1.

Table 1 Conventional generator parameters of the modified IEEE RTS-79 system

$P_G^{\max}$ (MW)	$P_G^{\min}$ (MW)	R_G^{DA} (MW/h)	c_G (\$/MW)	d_G (\$/MW)
50	10	15	55	16
76	15.2	22.8	50	18
100	20	30	48	19
155	31	46.5	45	20
197	39.4	59.1	42	21
350	70	105	40	22
400	80	120	40	23

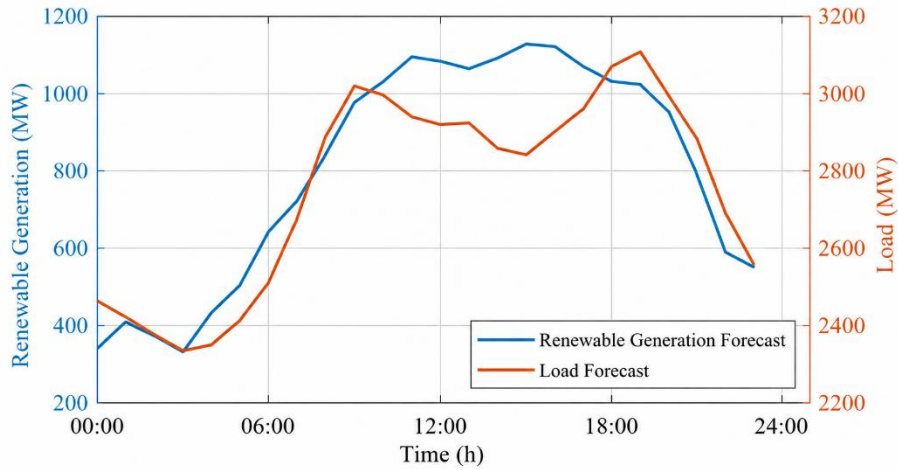


Fig.1 Day-ahead load and renewable generation forecast curves

2. IEEE 118-bus system

The IEEE 118-bus system consists of 118 buses, 186 transmission lines, and 54 generating units. All generating units with capacities of 400 MW or above are replaced by wind farms.

Generator outage states are enumerated up to the third order, while transmission-line outage states and combined outage states are enumerated up to the second order. The initial forced outage rates of generators and transmission lines are set to 0.01 and 0.001, respectively. The thermal capacity of every transmission line is assumed to be 500 MW. The parameters of conventional generating units are listed in Table 2. The forecasted load demand and renewable generation profiles are set to **1.5 times** those shown in Fig. 1. All other parameters are identical to those used in the IEEE RTS-79 system.

Table 2 Conventional generator parameters of the modified IEEE 118-bus system

$P_G^{\max}$	$P_G^{\min}$	R_G^{DA}	c_G	d_G	$P_G^{\max}$	$P_G^{\min}$	R_G^{DA}	c_G	d_G
(MW)	(MW)	(MW/h)	(\$/MW)	(\$/MW)	(MW)	(MW)	(MW/h)	(\$/MW)	(\$/MW)
100	20	80	52	25	185	37	148	50	22.5
104	20.8	83.2	52	25	255	51	204	50	22.5
107	21.4	85.6	52	25	260	52	208	50	22.5
119	23.8	95.2	52	25	304	60.8	243.2	45	20
136	27.2	108.8	52	25	320	64	256	45	20
140	28	112	52	25	352	70.4	281.6	45	20
148	29.6	118.4	52	25					